GREEN AI: Making AI More Sustainable

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AI's Unnoticed Environmental Impact

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AI has been rapidly changing our world and how we interact with it. Whether it is scientists predicting advanced protein folds, small businesses deploying AI websites with AWS, or customer service applying chatbots for customer interaction, nearly everyone is interacting with artificial intelligence in some form. Machine learning has become one of the most popular specializations for computer science developers.

Despite AI's international ubiquity among hundreds of millions of users, most are unfortunately unfamiliar with terms like "Green AI" or "carbon-aware computing." Sustainability rarely surfaces in discussions about software architecture or model development even though incorporating sustainable practices into software engineering is crucial, especially looking at the environmental cost of these large GenAI models.

Running these massive Generative AI models requires a significant amount of energy. AI queries require ten times the energy of a regular Google search. Training the BLOOM Generative AI model resulted in 50 tonnes of greenhouse gas emissions total. OpenAi's training of GPT-3, one of the smaller models, produced 502 total tonnes of emissions. More modern, complex models required even more. As our models grow, the pollution required to train and run these algorithms will balloon in size; it is projected that 30% of total worldwide energy usage will be from AI. Not only do running and developing these models require exorbitant amounts of pollution to be released, but a large amount of clean water is required to properly cool and maintain the hyperactive GPU's and TPU's of the machines running these models. As more people use AI and more complicated models are being trained, the cost to the environment and our resources will grow in magnitudes.

The term "Green AI" connects two key aspects: sustainability and artificial intelligence. On one hand, it focuses on how AI can be applied to improve sustainable practices. Examples include developing predictive models for environmental change, forecasting water usage in agriculture, and supporting wildlife conservation efforts.

On the other hand, it looks at how to make the training and development of AI models themselves more sustainable. This includes monitoring carbon footprints, optimizing hyperparameter tuning, and designing more efficient algorithms.

This paper will highlight the sustainability costs of AI models and explore ways to incentivize environmentally conscientious choices, making terms like Green AI and carbon-aware computing more prominent in the Big Data community, and leading to actionable outcomes that promote environmental responsibility.

Sustainable AI Practices and Solutions

As AI has such a pronounced environmental impact, it's important to find strategies to reduce this footprint. This, of course, also means that both companies and AI users need to prioritize environmental responsibility; methods for this will be explored in later sections. But before that, it is useful to examine some general approaches to reducing the environmental impact of AI systems.

For example, many large-scale companies deploy AI models on AWS or Microsoft Azure, both of which operate data centers across multiple regions worldwide. Each data center has a different Power Usage Effectiveness (PUE) value due to variations in cooling efficiency, climate, and infrastructure. For example, electricity in North Central Sweden is largely generated from low-carbon, renewable sources, making data centers in this region more sustainable to operate. In contrast, data centers located in warmer or less energy-efficient regions require significantly more energy for cooling, increasing their overall environmental impact. A PUE closer to 1 indicates greater energy efficiency; for instance, the BitFury data center in Georgia has a PUE of approximately 1.02, while Amazon's most efficient regions, such as those in Europe, have reported PUE values around 1.04. Being intentional about which data centers are selected from a sustainability perspective and incentivizing this type of decision-making would be an ideal first step toward making AI deployments more carbon-efficient.

Additionally, hardware optimization techniques can be employed. One option could be opting for specialized energy-efficient process units with reduced-complexity instruction sets made specifically for machine learning operations, which will minimize the instruction

translation overhead. Some examples are Tensor Processing Units (TPUs), Neural Processing Units (NPUs), or Graphics Processing Units (GPUs) with Tensor Cores.

Another important factor in improving AI efficiency is the use of smaller, task-appropriate models instead of heavily overparameterized architectures, which often incur substantial energy costs while providing only marginal gains in performance. For example, many classification or information extraction tasks can be effectively handled by fine-tuned lightweight transformer models such as DistilBERT or ALBERT, rather than large-scale models with billions of parameters, achieving comparable accuracy while requiring significantly less compute and energy during both training and inference. Expanding on this, when using a virtual server to deploy models in the cloud, selecting instance types that are scaled for the necessary workload, rather than overproviding with a high-performing instance type, can further reduce unnecessary energy consumption.

Gamifying Sustainability for Awareness and Adoption

Increasing AI users' awareness of their environmental impact, guiding more efficient AI usage, and actively incentivizing the implementation of sustainable techniques will help businesses and users prioritize sustainability, rather than treating

Defining for users their specific environmental impact, providing tailor-made strategies to further reduce carbon usage, and actively incentivizing the implementation of sustainable techniques will help businesses and users prioritize sustainability, rather than treating the environmental effects as an afterthought, if they are considered at all.

The first step is improving monitoring techniques. LLM API usage does provide information on token usage, but it would be helpful to also have LLM APIs return environmental impact data as part of their responses, alongside standardized calculations of carbon emissions, water usage, and electricity consumption across model infrastructure. By attaching quantifiable metrics to these impacts, users gain a concrete understanding of their exact footprint, replacing vague or abstract guilt with actionable, measurable insight.

Of course, simply looking at the carbon footprint of AI usage on its own can be misleading, since different users rely on AI for very different purposes. For example, one user might use AI to better understand lecture notes but do so inefficiently—by inputting excessive

text to summarize or repeatedly re-asking similar questions. While this may reflect inefficient energy use, the overall carbon footprint is still relatively small compared to someone developing a sophisticated AI model to predict potential healthcare treatment side effects before a drug enters clinical trials.

What matters is not just measuring carbon consumption, but using that data to identify areas where efficiency can be improved. By highlighting these inefficiencies, we can incentivize users to adopt more sustainable and energy-conscious practices.

There are a few ways the user can be presented with how they are misusing their resources, which produces unnecessary emissions. For example, imagine an AWS dashboard that highlights underutilized resources, like an EC2 instance that isn't being used effectively. By right-sizing or downsizing that instance, the user could gain sustainability points, rewarding optimization, and encouraging more efficient resource usage. Also, points can be accrued by using more sustainable hardware that is known for efficiency when it comes to large ML workloads.

Users earn points that function like a credit reward system, similar to a cashback credit card. These points can be redeemed for billing credits toward AI usage costs or even converted into monetary cashback. In addition, users, projects, and companies can receive awards and badges for sustainable practices, representing recognition for implementing efficient, low-impact AI workflows. These badges signal employees and teams with strong knowledge of responsible AI operations.

Now for clarification, in rewarding users in sustainability, we want to have a comparison of expected energy usage for the problem at hand, and then compare to the users actual usage of energy. This will require really indepth methods of calculating averages. Of course, studies could be done for different AI use cases to measure how much average users with similar use cases end up producing emissions-wise, there could also be AI platforms (like AWS) that incentivizes users to deploy and run AI models through their services, and within these services monitoring can be done for very similar use cases, and calculate the average expected emissions from such a use case.

By comparing projects based on their AI awareness awards, organizations can quickly identify high-performing, sustainability-mature initiatives and highlight weaker areas that still need development. On a broader scale, company-wide comparisons of environmental awareness indicate which organizations are leading in responsible AI adoption and which may require

guidance, new policies, or technical improvements to reduce resource waste and operational emissions.

Case Studies and Benchmarks

Currently there are few products ready on the market that are built to gamify sustainability for AI users. Although there are a few initiatives that incorporate aspects of this goal that can be discussed.

Several tools have been developed to help monitor users' sustainability impacts. One example is AWS's Customer Carbon Footprint Tool (CCFT), which provides visibility into emissions associated with cloud usage.

These emissions are typically divided into two categories. The first is Scope 2, which refers to indirect emissions from the electricity used to power services such as EC2 instances and S3 storage. The second is Scope 3, which includes emissions associated with the manufacturing, transportation, installation, and end-of-life treatment of the hardware that supports those cloud services.

Together, these categories offer a more complete view of the environmental impact of using cloud infrastructure. Of course AWS's footprint tool has some major limitations, namely that it aggregates usage data rather than displaying specific workload data. But, the tool overall is a major step in the right direction, as it attempts to calculate the predicted emissions and the actual emissions consumed, and as AWS is used by many large scale companies already, these kinds of integrated tools can really aid in different user's and companies overall AI impact.

A look at the gamification aspect; several recycling "money back" types of structures have been implemented where recycling plastic has earned back profits on households to try to encourage users to throw plastic in recycling bins rather than trash bins. In Singapore there are essentially reverse vending machines, specifically made to pay back for deposition plastic bottles.

Other similar systems exist called Deposit-return schemes; these programs essentially have a deposit in purchasing drinks that can be returned after the empty container is returned. Some countries that have this system implemented include Germany, Norway, and the United States, and these systems lead to large scale participation with bottle return rates of 80-90%

A major avenue for developing effective sustainability and AI gamification solutions is through startups, where different approaches can be tested and refined.

Several AI startup initiatives are helping increase awareness of individual and organizational carbon contributions. For example, CO2 AI aims to automate the tracking and analysis of carbon emissions for companies. Other startups with similar goals in emissions tracking include Watershed, Climatiq, and Klima, all of which focus on helping organizations measure, manage, and reduce their environmental impact. For actual gamification there is a newer app called AWorld that rewards points for everyday environmental choices like using public transportation, plant-based meals etc; tracking carbon footprint and rewarding environmental behaviors.

Policy

Although AI can involve significant energy use and carbon emissions, projects leveraging AI have the potential to preserve nature and, over the long term, actually reduce overall environmental impact.

Artificial intelligence (AI) has the potential to significantly support environmental conservation and climate action. Tools like Google Earth AI leverage satellite imagery and machine learning to monitor forests, oceans, and wildlife habitats in near real-time. By analyzing vast amounts of data, AI can detect deforestation, illegal fishing, or pollution events much faster than human teams could. This enables governments and organizations to respond quickly and implement targeted conservation strategies, reducing the environmental impact of human activity.

Beyond monitoring, AI can optimize energy consumption and resource management. For instance, AI-driven systems can improve electricity grids by predicting energy demand, reducing waste, and integrating renewable sources like solar and wind more efficiently. In agriculture, AI helps farmers use water, fertilizers, and pesticides more precisely, minimizing runoff and soil

degradation while maintaining high crop yields. Over time, these efficiencies can lead to a net reduction in carbon emissions, offsetting some of the energy cost associated with AI itself.

On the other hand, not using AI—or delaying its adoption—can have negative environmental consequences. Without AI-powered monitoring, large-scale environmental changes such as illegal logging, habitat loss, and ocean pollution may go unnoticed until irreversible damage occurs. Inefficiencies in energy systems, agriculture, and transportation continue unchecked, resulting in higher carbon emissions and greater strain on natural resources. Essentially, the absence of AI could mean slower responses to urgent environmental threats, leading to a larger ecological footprint over time.

That being said, despite the potential benefits of AI for the environment, it is crucial that users understand how AI itself impacts the planet, and that strong support is provided to sustainability initiatives. Transparency should be a core principle: companies developing AI models must clearly report the energy consumption and carbon footprint of their systems. Currently, one of the major barriers in monitoring environmental impact is the lack of standardized reporting, making it difficult to assess and reduce AI's ecological costs.

In addition to transparency, policies should encourage innovation in environmentally responsible AI. Startups and organizations that develop AI tools specifically aimed at sustainability—such as energy-efficient models, AI for conservation, or climate prediction—should receive funding and support. By combining clear reporting requirements with incentives for green AI, we can ensure that AI not only avoids harming the environment but actively contributes to its preservation.

Call to Action and Final Thoughts

With the bloom of AI usage, sustainability should be incorporated into the strategy governing how models are trained, deployed, and used. However, the focus in AI development is, at times, placed on improving metrics such as accuracy, throughput, and benchmark performance, to the extent that sustainability-related considerations are often overlooked or not widely understood. Increasing awareness of Green AI, alongside the development of systems and

frameworks that transparently quantify the environmental impact of AI usage, can help users better understand their individual contribution and the broader implications of AI at scale. By making these impacts more visible, sustainability can be elevated to a core priority rather than an afterthought.

Citations